Encouragement as Pedagogy: Using AI-Mediated Mathematical Participation to Address Math Anxiety in Education and Outreach

Soumya Banerjee
University of York
United Kingdom
soumya.banerjee@york.ac.uk

Abstract

Math anxiety, a persistent apprehension about mathematical tasks that measurably degrades working memory and performance, affects a large share of students and adults and leads many to avoid mathematics entirely. We argue that the same division of labor that allows a non-mathematician to contribute to frontier mathematical research, supplying framing and encouragement while an AI system supplies technical execution, offers a promising template for math education and outreach aimed at anxious learners. Rather than asking an anxious student to close a knowledge gap before engaging with mathematics, this model lets them engage first, in a low-stakes register (asking questions, expressing uncertainty, requesting another attempt) while an AI tutor absorbs the technical load and a human retains responsibility for pacing, tone, and correctness. We connect this argument to two independent evidence bases: research on math anxiety and avoidance, and recent randomized evidence that supervised generative AI tutoring can match or exceed human one-to-one tutoring on mathematics learning outcomes. We argue that the encouragement-only mode of participation documented in AI-assisted mathematical research is not a curiosity restricted to elite problems, but a design pattern with direct application to classrooms and public outreach, and we outline the safeguards this pattern requires, particularly continued human oversight of tone, pacing, and factual accuracy.

1 Introduction

Math anxiety is not simply a dislike of mathematics. It is a measurable apprehension that interferes with the working memory resources a person needs to actually do mathematics, producing a self-reinforcing cycle in which anxiety degrades performance, poor performance increases anxiety, and both drive long-term avoidance of math coursework, math-intensive majors, and math-intensive careers [Ashcraft, 2002]. This avoidance pattern is well documented and appears early, with evidence that math anxiety can emerge in the first years of schooling and remain stable into adulthood [Hembree, 1990]. A large meta-analytic literature confirms that anxiety and achievement are reciprocally linked, and that relief strategies which reduce anxiety without addressing avoidance tend not to produce durable gains [Hembree, 1990].

At the same time, a well-documented recent case in AI-assisted mathematical research showed that a non-mathematician's contribution to a genuine mathematical result was limited almost entirely to messages of encouragement, an interaction pattern that required no mathematical competence at all [Anthropic, 2026]. This raises a natural question for education and outreach: if encouragement and framing are sufficient for a non-expert to meaningfully participate in frontier mathematical research,

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: The 6th Workshop on Mathematical Reasoning and AI.

could a similar low-stakes mode of participation help precisely the population that mathematics currently repels, namely anxious learners who avoid engaging with mathematical content at all? We argue that it can, and that the mechanism is not a metaphorical extension but a fairly literal one: an AI tutor capable of absorbing technical difficulty while a human, whether a teacher, parent, or the learner themselves, supplies persistence and reassurance, changes the entry cost of engaging with mathematics in the same way it changed the entry cost of contributing to open mathematical problems.

This paper connects two literatures that are rarely discussed together, math anxiety research and recent randomized evidence on AI tutoring efficacy, and argues that their combination points toward a specific design pattern for math education and outreach.

2 Math Anxiety as an Avoidance Problem

The central practical consequence of math anxiety is avoidance. Highly math-anxious individuals avoid elective mathematics coursework, avoid college majors that emphasize mathematics, and avoid career paths that involve mathematics, a pattern that compounds over a lifetime and forecloses options well beyond the classroom [Ashcraft, 2002]. Because the anxiety response itself competes for working memory resources needed to perform mathematical tasks, math-anxious individuals often perform worse than their underlying competence would predict, particularly under timed or evaluative conditions [Ashcraft, 2002, Ashcraft and Kirk, 2001]. This creates a difficult bind for conventional instruction: the settings that are best suited to building competence (timed practice, cold calling, graded problem sets) are frequently the settings that are most aversive to anxious learners, so the people who would benefit most from additional practice are the people least likely to seek it out voluntarily.

Outreach programs face a related but distinct version of this problem. A member of the public who was made anxious by mathematics in school has little incentive to attend a public lecture on an open mathematical problem, read a popular article about a new result, or ask a question at a science festival, even when the underlying content might genuinely interest them. The barrier is not access to information; it is the anticipated experience of being asked to perform, or of being expected to already understand.

3 A Low-Stakes Entry Point: Encouragement Without Expertise

The interaction pattern documented in AI-assisted mathematical research offers a way to reframe this barrier. In that case, a non-mathematician's role was not to demonstrate competence but to sustain engagement through short, low-stakes messages, variants of "keep going" and "believe in yourself," while an AI system carried the technical burden and an initial period of failure. What made this mode of participation viable was that it required no performance from the human at any point: no correct answer, no demonstrated understanding, nothing that could itself trigger anxiety.

We argue that the same structure maps directly onto tutoring for an anxious learner. Rather than requiring a student to produce a correct answer under evaluative pressure, an AI tutor can be structured so that the student's primary required input is engagement itself, asking a question, saying they are stuck, or requesting another attempt, while the tutor absorbs the technical explanation and a supervising human (a teacher, in current deployments) retains responsibility for calibrating tone and pacing. This reframes the learner's role from performer to participant, which is precisely the shift that made non-expert participation possible in the research case.

This is not a hypothetical extrapolation. A recent randomized controlled trial integrating a pedagogically fine-tuned AI tutor, supervised in real time by expert human tutors, into secondary school mathematics instruction found that AI-supported tutoring sessions were at least as effective as sessions with a human tutor alone across every learning outcome measured, and produced modestly higher rates of knowledge transfer to subsequent topics [LearnLM Team, Google and Eedi, 2025]. Notably, the supervising tutors in that trial reported that the AI system consistently generated Socratic, encouraging dialogue, and that their own interventions were concentrated on moderating pacing and adding social and emotional nuance rather than correcting technical content [LearnLM Team, Google and Eedi, 2025]. This division of labor, an AI system providing patient, non-judgmental technical scaffolding and a human providing emotional calibration, mirrors the encouragement-only pattern

from the research case, but applied at the scale of routine instruction rather than a single celebrated result.

This also fits within a broader empirical case for tutoring generally. One-to-one tutoring has long been recognized as an unusually effective form of instruction, with foundational work suggesting gains on the order of two standard deviations over conventional classroom teaching under ideal conditions, and more conservative subsequent meta-analyses still finding substantial, replicable benefits from tutoring across contexts [Bloom, 1984]. The persistent obstacle has always been cost and scale, since one-to-one human tutoring cannot be delivered to every anxious student in every classroom, and even more conservative meta-analytic estimates of tutoring's average effect remain far larger than most classroom-level interventions achieve [Nickow et al., 2020]. AI tutoring does not resolve every aspect of that scaling problem, but the randomized evidence above suggests it can extend a meaningful share of tutoring's benefits, including its emotionally patient, non-judgmental register, considerably further than human tutors alone can currently reach.

4 Implications for Outreach

The same design pattern extends beyond formal instruction to public mathematical outreach. Museums, science festivals, and popular science writing have long struggled to engage audiences who consider themselves "bad at math," often by simplifying content until little of the original mathematics remains. An alternative suggested by our argument is to preserve the technical depth of the content while changing what is asked of the participant: instead of asking a visitor to solve a problem, an outreach exhibit or public-facing tool built on an AI system could invite them to pose a question, express curiosity about why a result matters, or simply encourage an AI agent working through a real problem in real time, much as the non-mathematician in the research case did. This lets a general audience engage authentically with frontier mathematics without the performance anxiety that a request to solve, prove, or answer would trigger. Terence Tao's proposed notion of "citizen mathematics," in which non-experts contribute modular, low-stakes pieces to larger collaborative projects, provides a template for how such outreach could be structured beyond a single interaction [Tao, 2026].

5 Risks and Necessary Safeguards

This design pattern is not risk-free, and the randomized tutoring evidence itself points to where the risks lie. First, an AI tutor operating without supervision can misjudge pacing in ways that frustrate rather than reassure a student; in the reported trial, the single most common reason supervising tutors intervened was to prevent an AI system's persistent Socratic questioning from exasperating a student who had already understood the material [LearnLM Team, Google and Eedi, 2025]. For an anxious learner, this failure mode is especially costly, since frustration is precisely the experience the intervention is meant to avoid. Second, factual errors, though rare in the reported trial, are not eliminated by encouragement, and an anxious learner may be especially unlikely to question a confident but incorrect explanation, making human oversight of accuracy essential rather than optional [LearnLM Team, Google and Eedi, 2025]. Third, there is a risk of mistaking comfort for competence: an intervention that reduces anxiety in the moment but never builds genuine mathematical understanding would only postpone avoidance rather than resolve it, and evaluation of these tools should measure durable learning transfer, not just session-level satisfaction. Any deployment of this pattern in education or outreach should retain a human, whether a teacher, parent, or facilitator, with the ability to review, edit, and moderate the AI system's interactions, mirroring the supervised structure used in the randomized trial rather than an unsupervised deployment.

6 Conclusion

Math anxiety produces avoidance, and avoidance is difficult to treat with interventions that themselves demand performance from an anxious learner. The same interaction pattern that allowed a non-mathematician to meaningfully contribute to a frontier mathematical result, sustained engagement through encouragement rather than demonstrated expertise, offers a workable template for reaching anxious learners in both formal education and public outreach. Randomized evidence on supervised AI tutoring already shows that this pattern can match or exceed human one-to-one tutoring on real

learning outcomes, provided a human remains responsible for pacing, tone, and accuracy. We think research and practice on math anxiety should treat this encouragement-centered mode of engagement as a design target in its own right, not merely a byproduct of capable AI systems, and we encourage the workshop community to study how far this pattern generalizes across ages, settings, and levels of anxiety.

References

Anthropic. Learning more about Claude's mathematical capabilities. Anthropic Research blog, 2026. URL `https://www.anthropic.com/research/riemann-zeta`. Accessed September 2026.

Mark H. Ashcraft. Math anxiety: Personal, educational, and cognitive consequences. *Current Directions in Psychological Science*, 11(5):181–185, 2002. doi: 10.1111/1467-8721.00196.

Mark H. Ashcraft and Elizabeth P. Kirk. The relationships among working memory, math anxiety, and performance. *Journal of Experimental Psychology: General*, 130(2):224–237, 2001. doi: 10.1037/0096-3445.130.2.224.

Benjamin S. Bloom. The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational Researcher*, 13(6):4–16, 1984. doi: 10.3102/0013189X013006004.

Ray Hembree. The nature, effects, and relief of mathematics anxiety. *Journal for Research in Mathematics Education*, 21(1):33–46, 1990. doi: 10.2307/749455.

LearnLM Team, Google and Eedi. AI tutoring can safely and effectively support students: An exploratory RCT in UK classrooms. arXiv preprint arXiv:2512.23633, 2025. URL `https://arxiv.org/pdf/2512.23633`.

Andre Nickow, Philip Oreopoulos, and Vincent Quan. The impressive effects of tutoring on PreK-12 learning: A systematic review and meta-analysis of the experimental evidence. NBER Working Paper, 2020. URL `https://www.nber.org/papers/w27476`.

Terence Tao. Terence Tao on AI: A living summary. Personal website, 2026. URL `https://teorth.github.io/tao-web/ai-views.html`. Accessed September 2026.